



DELPHI GROUP

delphi.cmu.edu

Carnegie Mellon University

UC Berkeley

The University of
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MICOM

Michigan Public Health Integrated Center for Outbreak Analytics and Modeling

micom-insight.net

Give this **worksheet**
a try to test our tools!

Or review the
solution examples.



Epidata Repository

Delphi curates data streams from industry and STLT partners, providing a discoverable library of disease surveillance data



EpiPortal —
Discover
indicators
with the data
library

Delphi Epidata API —
Programmatic access to
Delphi-hosted data with:

epidatr: the R API client

epidatpy: the Python API client

R Tools for Epidata

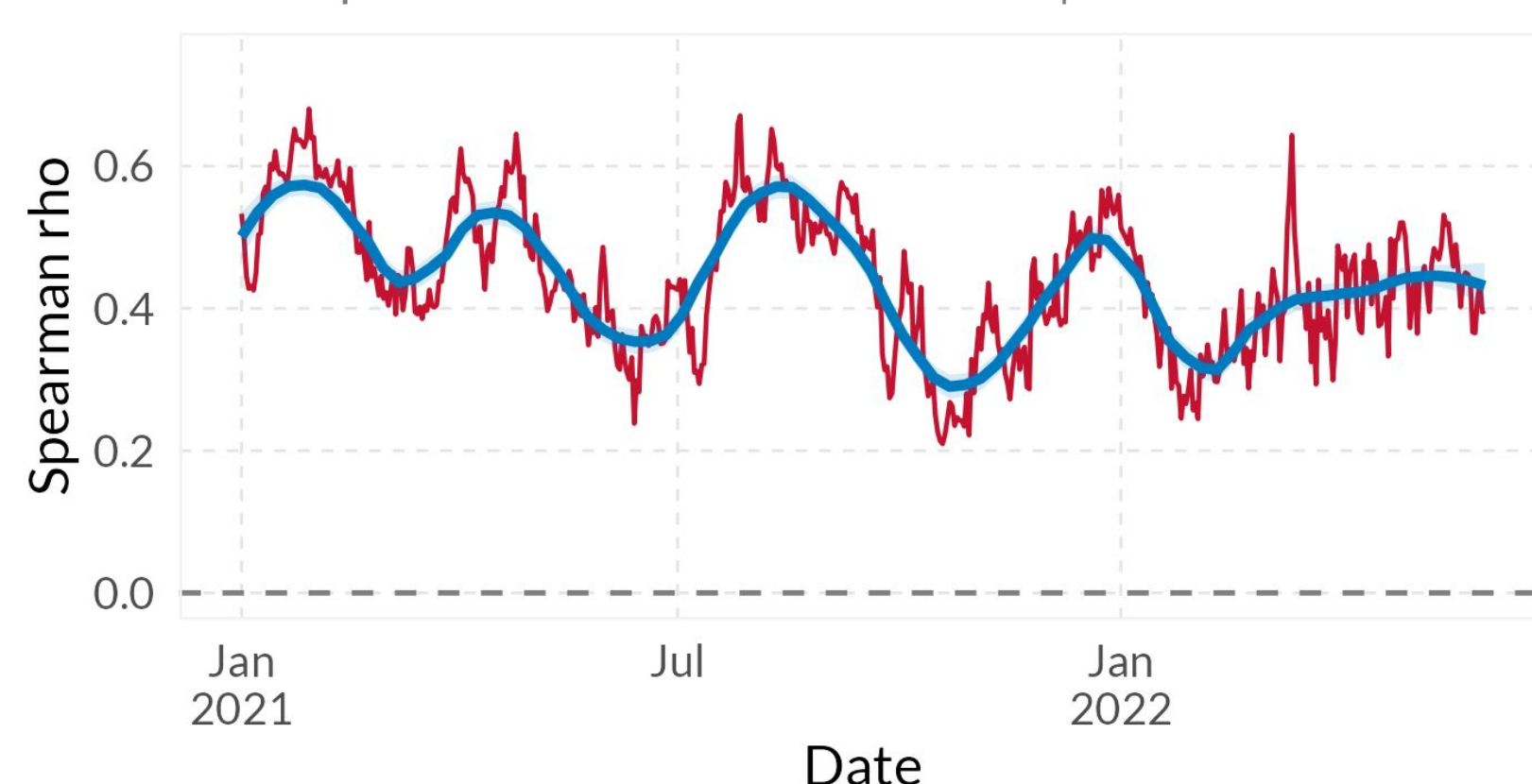
epiprocess — Visualize, summarize, transform epidemic time series and reporting patterns

Detect outliers, compute growth rates, estimate correlations across groups, and produce data revision reports.

epipredict — Generate statistical forecasts quickly; zero customization required, but many options & custom framework available

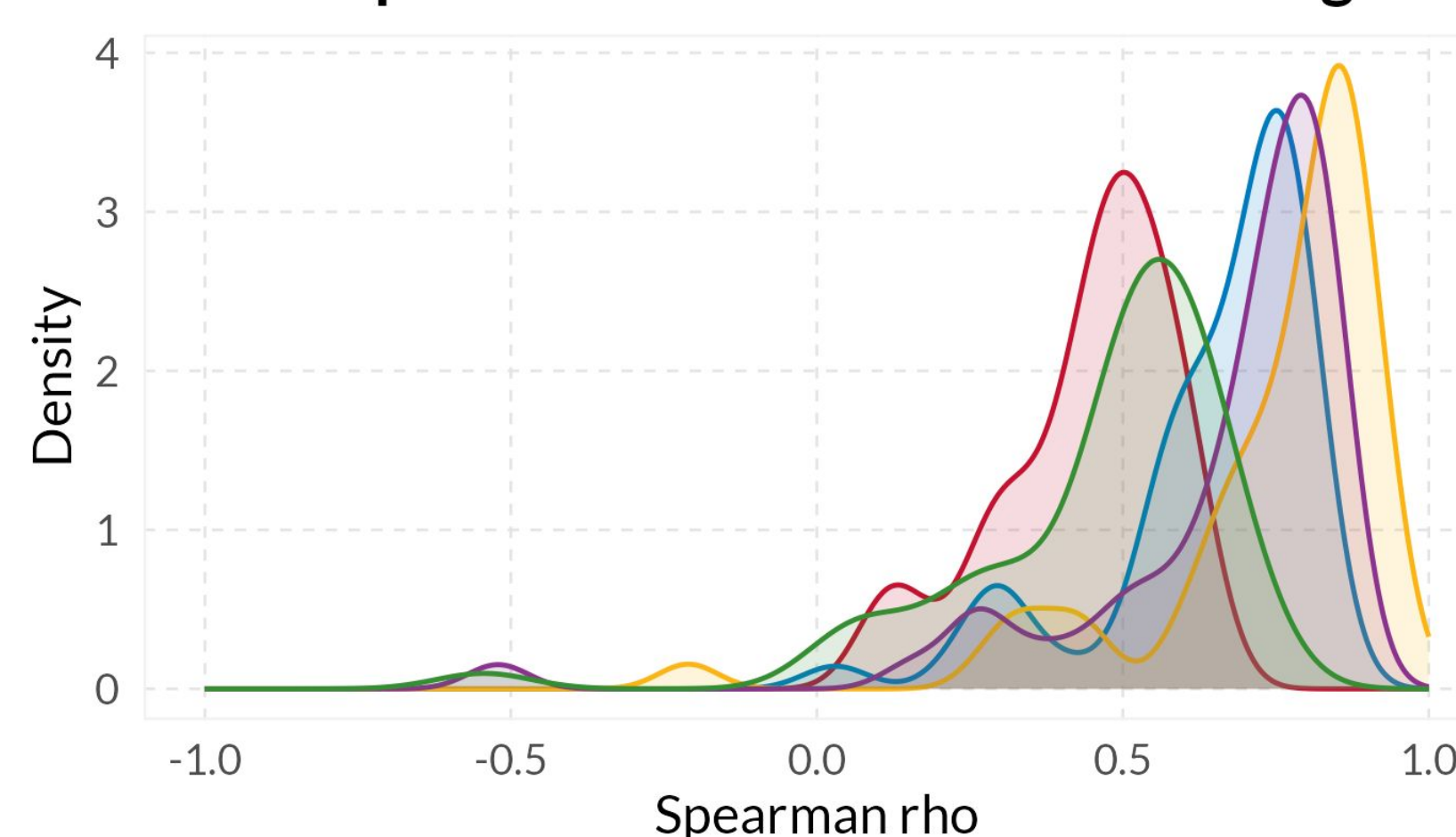
Correlation over time

Spearman rho across locations | Mean = 0.442



Trend — Correlation — LOESS smoothed trend

Temporal correlations at different lags



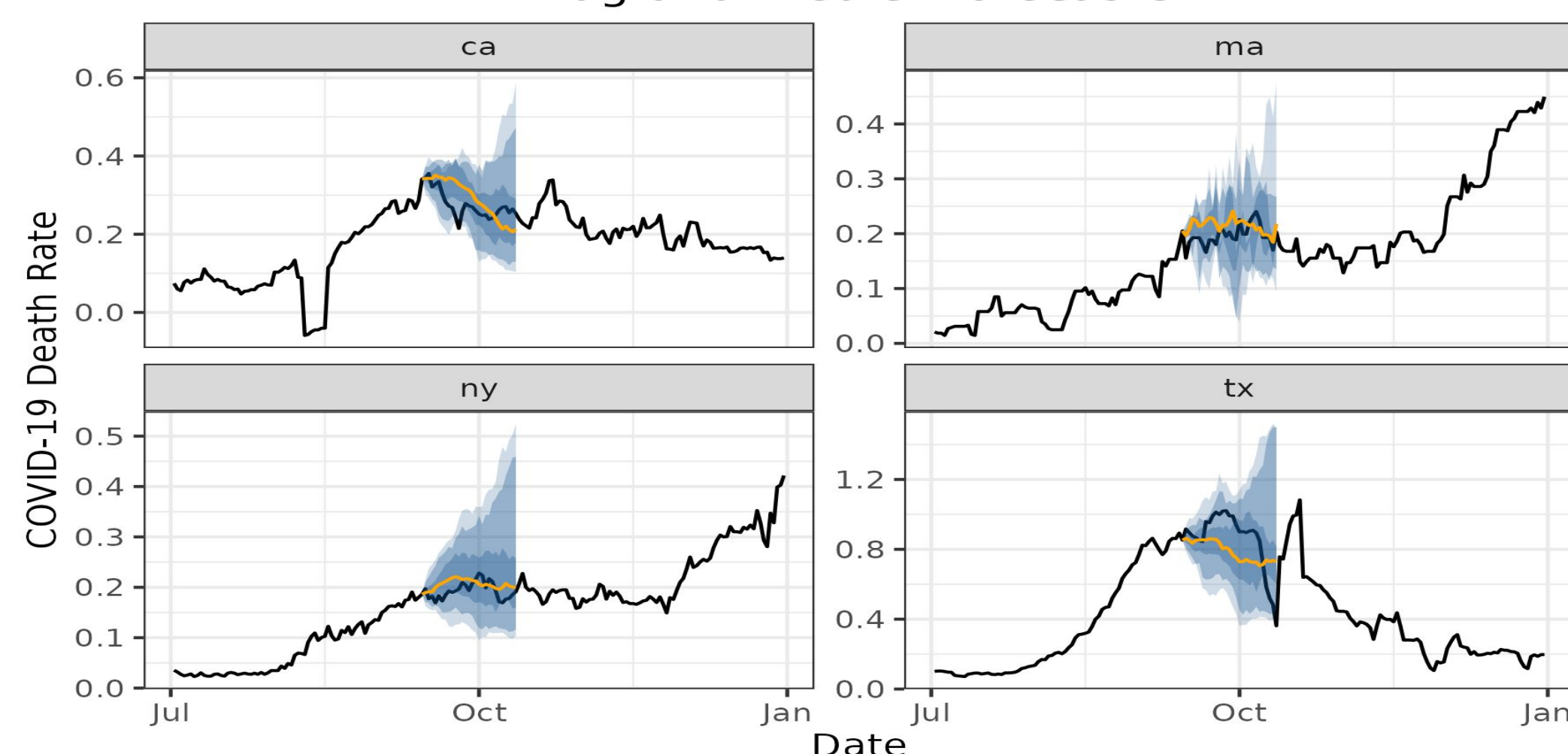
Lag (days) — -28 — -14 — 0 — 14 — 28

This + more is in Delphi's new
correlation analysis notebook
tool; see this example:



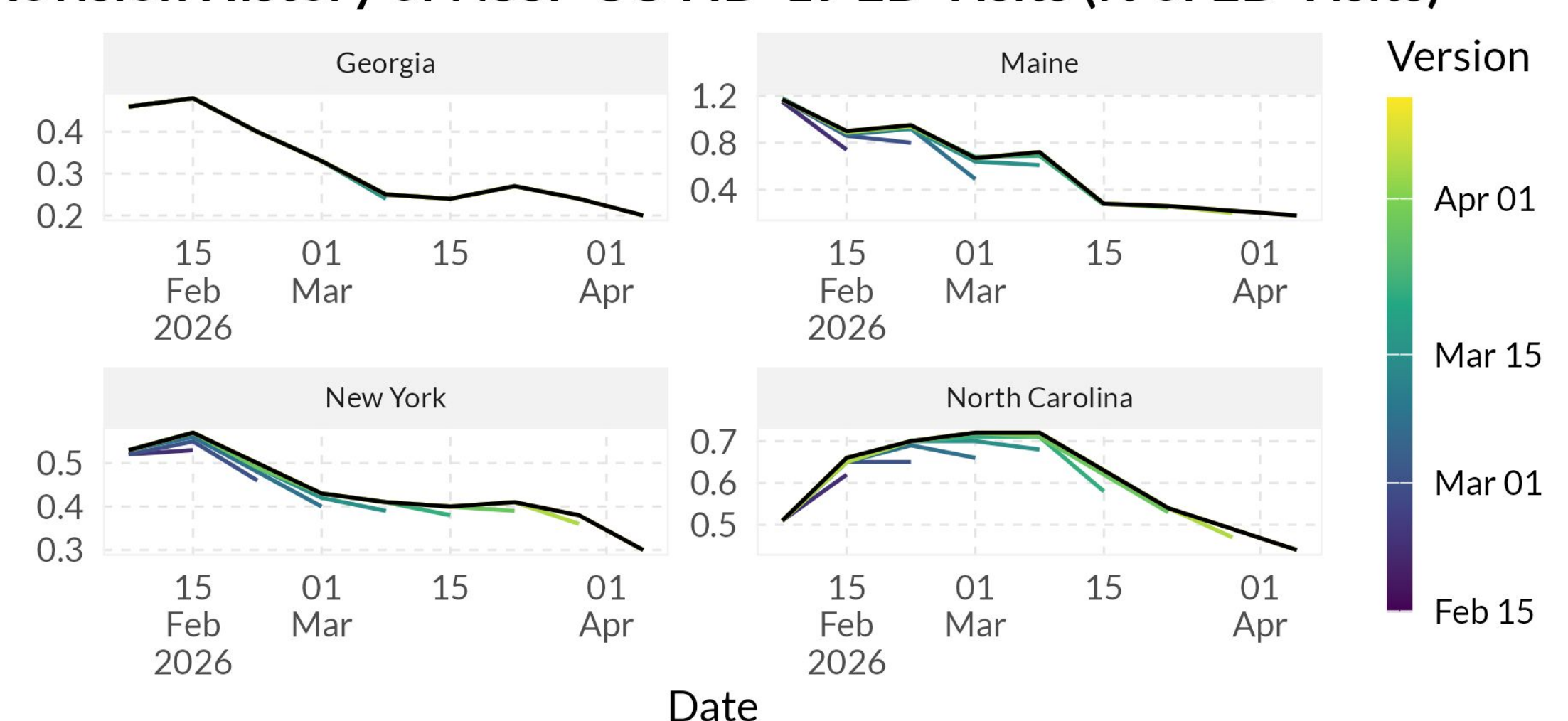
Correlation tools. Epiprocess offers simple and efficient features for calculating lagged correlations between two signals across locations or across time.

Plug-and-Predict Forecaster



Plug-and-predict forecasting. Quickly generate forecasts using models tuned for CDC & CFA forecasting challenges.

Revision History of NSSP COVID-19 ED Visits (% of ED Visits)



Archive plotting. Convenient autoplot method to visualize data revisions.



Tool overview!

Recent R Package Updates

epiprocess

- Interactive time series plotting
- Support for creating a full version history from a line list
- Enhanced data set summary

epipredict

- New plug-and-predict "climatological" baseline model. Matches data from previous years to current season phase to make predictions.

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